

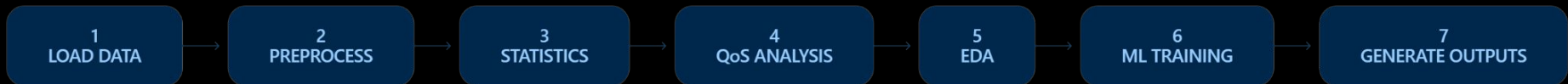
REVIEW 2

Mobile Network Quality-of-Service Analysis and Prediction

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This review focus on:

Understand Data → Check Quality → Prepare Data → Analyze Statistics → Explore Relationships

- **Dataset Understanding** — structure, features and basic statistics
- **Data Quality** — missing values and outlier identification
- **Data Preparation** — missing-value treatment and feature scaling
- **Statistical Analysis** — mean, variance, skewness and kurtosis
- **Exploratory Data Analysis** — univariate, bivariate and multivariate patterns

Outlier analysis

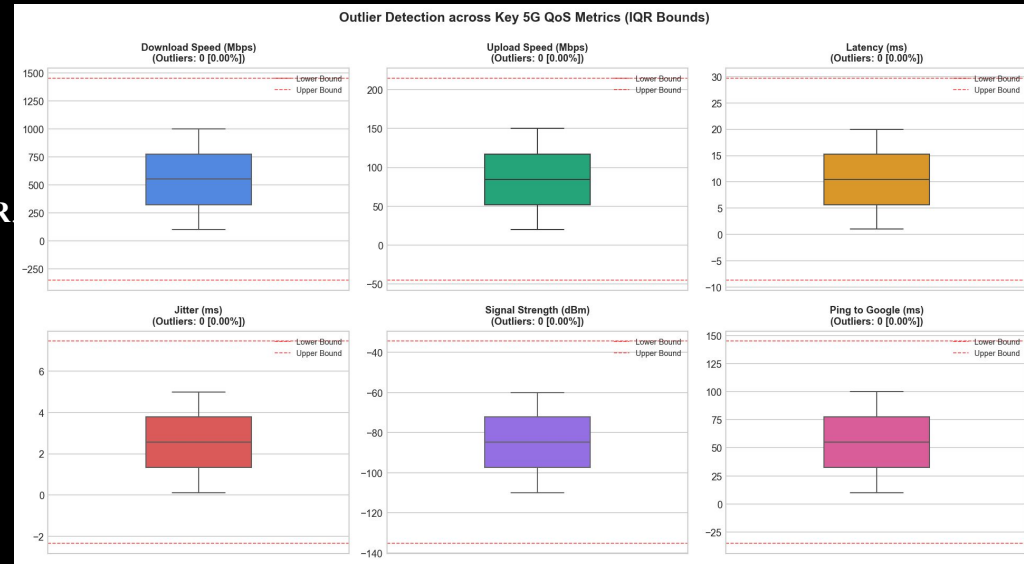
Method

- Used **Tukey's IQR method** to identify statistical outliers.
- For each numerical feature, calculated **Q1, Q3 and IQR**.
- Outlier limits were defined as **$Q1 - 1.5 \times IQR$** and **$Q3 + 1.5 \times IQR$** .

Key Results

- **12 numerical QoS/features analyzed**
- **0 outliers detected across all 12 features**
- **Therefore, 0% of the dataset was classified as outliers for these features.**

IQR analysis found no statistical outliers in the 12 numerical features, indicating that the dataset does not require outlier removal at this stage.



Dataset Inspection - head(), tail(), info(), describe()

Basic inspection established the dataset structure, feature types, completeness and statistical baseline before preprocessing and EDA.

```
inspection_data = {
    'shape': df.shape,
    'head': df.head(),
    'tail': df.tail(),
    'info': info_str,
    'describe': df.describe(),
    'dtypes_df': dtypes_df,
    'unique_df': unique_df,
    'numerical_cols': num_cols,
    'categorical_cols': cat_cols,
    'boolean_cols': bool_cols,
    'temporal_cols': ['Timestamp'] if 'Timestamp' in df.columns else []
}
```

Method / Output	Purpose	Outcome from Dataset
`shape`	Identifies dataset size	50,000 × 21
`head()`	Inspects first records	Confirms initial data structure
`tail()`	Inspects last records	Confirms end-of-dataset records
`info()`	Checks data types & completeness	21 features, no null entries
`describe()`	Provides statistical summary	Mean, Std, Min, Q1, Median, Q3, Max
`dtypes_df`	Categorizes column data types	Numerical, categorical, Boolean & temporal
`unique_df`	Checks unique values	Identifies categories/cardinality

Missing Value Treatment: Mean vs Median

```
df_mean_imputed = df.copy()
df_median_imputed = df.copy()

num_cols = df.select_dtypes(include=[np.number]).columns.tolist()
imputation_records = []
```

```
for col in num_cols:
    orig_missing = df[col].isnull().sum()
    mean_val = df[col].mean()
    median_val = df[col].median()
    std_val = df[col].std()
    skew_val = df[col].skew()

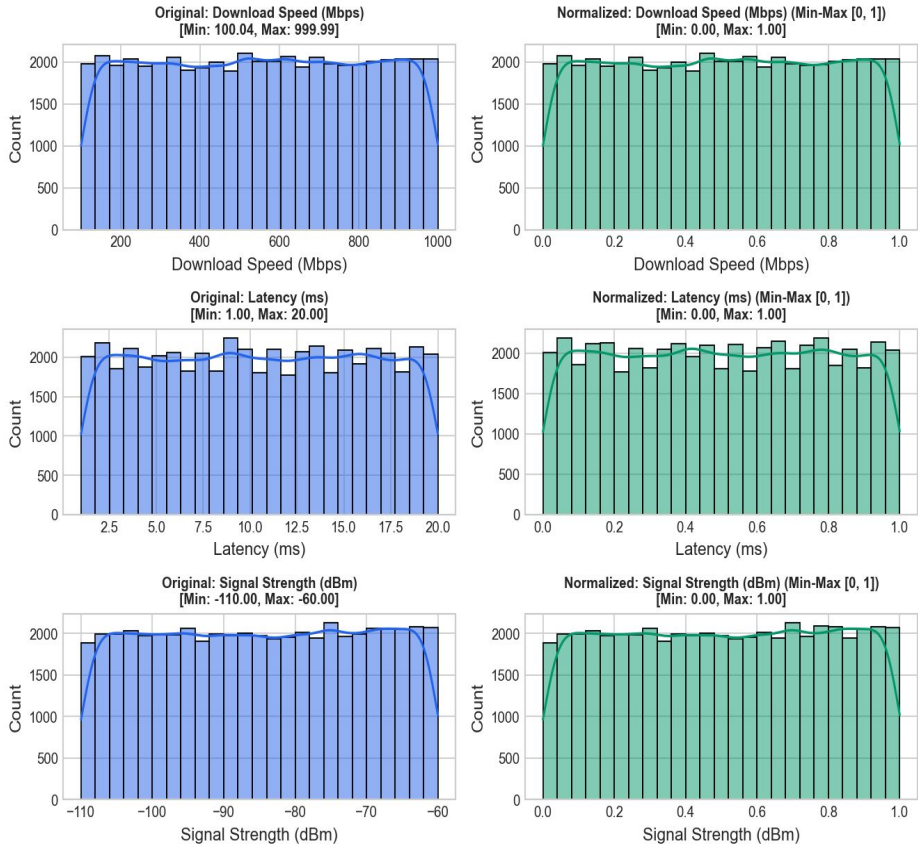
    df_mean_imputed[col] = df_mean_imputed[col].fillna(mean_val)
    df_median_imputed[col] = df_median_imputed[col].fillna(median_val)
```

```
if abs(skew_val) > 0.75:
    recommendation = "Median (Distribution is skewed / outlier sensitive)"
else:
    recommendation = "Mean (Distribution is relatively symmetric)"
```

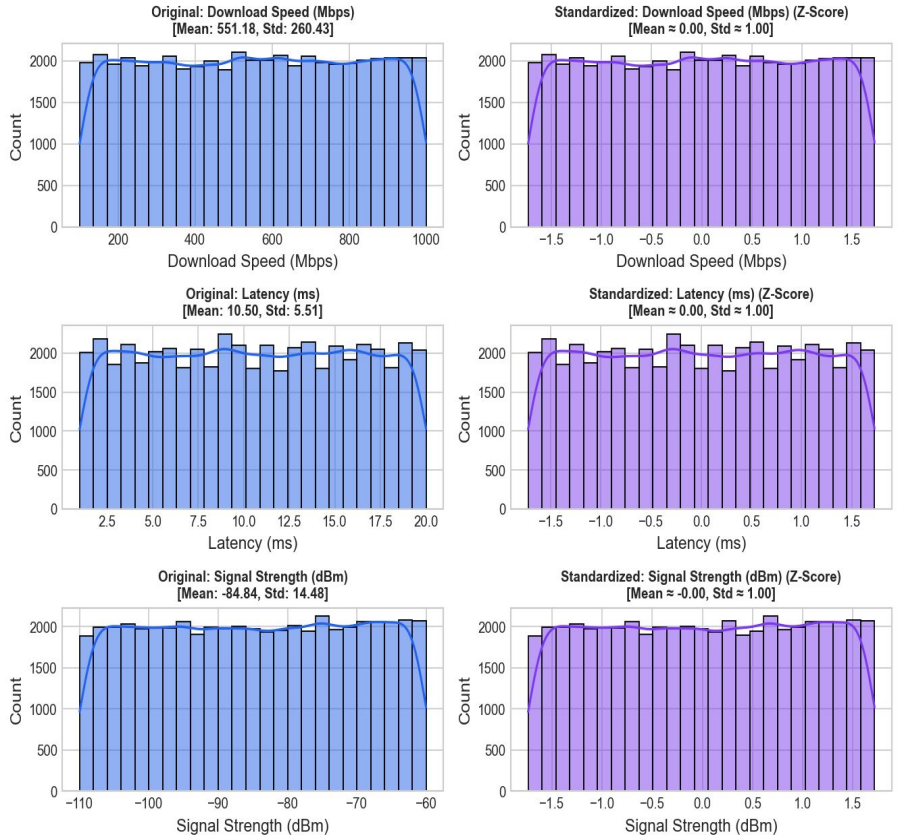
Step	Purpose	Outcome
df.copy()	Keeps original dataset unchanged	Original data preserved
mean()	Calculates column mean	Mean used as one imputation option
median()	Calculates column median	Median used as second option
fillna()	Replaces missing values	Missing values are filled
skew()	Checks distribution shape	Helps select Mean vs Median

Normalization and Standardization

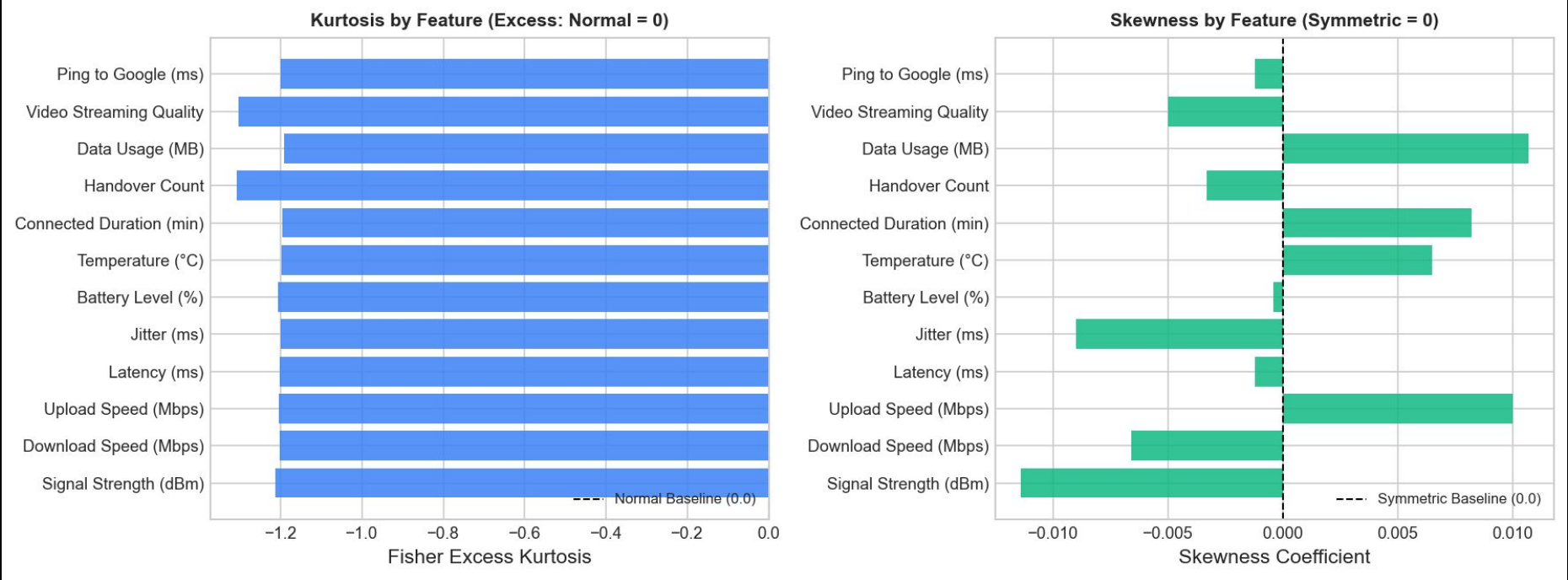
Min-Max Normalization Effect on Distribution Scale



Z-Score Standardization Effect on Central Tendency and Spread



Statistical Distribution Analysis ~Mean • Variance • Skewness • Kurtosis



THANKYOU